Robot Controller

Controller Series	IRCB501 Series	IRCB501 High-protection Series
Mounting mode	Vertical mounting, horizontal mounting, 19" rack mounting	Vertical mounting, horizontal mounting, rack mounting
Standard I/Os	16 inputs and 16 NPN outputs (extendable)	
Communication interfaces	Ethernet interface: Used for TCP/IP, Modbus TCP, Ethernet/IP, MC communication	
	EtherCAT-IN interface: EtherCAT salve –IN interface	
	EtherCAT-OUT interface: EtherCAT master –OUT interface	
	EtherCAT interface: Used for extension of external axes	
	RS232/RS485 interface: Used for serial and Modbus RTU communication (RS485 only)	
	USB2.0 interface: Used for backup/upload programs and export robot status information	
	Optional interface: Profinet slave , CC-LINK slave	
Control mode	PC programming platform control, teach pendant control, remote I/O control, remote Modbus control, and API control	
Power supply	Input voltage: Single-phase 200 VAC to 240 VAC, 10A/20A, 50 Hz to 60 Hz	Input voltage: Single-phase 200 VAC to 250 VAC, 23A, 50 Hz to 60 Hz
	Max. power consumption: 3.1 kW (depending on the robot model)	Max. power consumption: 4.5 kW (depending on the robot model)
IP rating	IP20	IP54 + anti metal dust
Operating conditions	Temperature: 5° C to 40° C ; Relative humidity: 20% to 95% RH@30° C (non-condensing)	Temperature: 0° C to 45° C; Relative humidity: 20% to 95% RH@30° C (non-condensing)
Dimensions	330mm×400mm×130mm	445mm×575mm×276mm
Weight	10kg	30kg
Applicable Robots	6-Axis: IR-R7H Series IR-R4H Series IR-R11 Series IR-R15H Series	6-Axis: IR-R10 Series, IR-R25 Series IR-R4H/R7H/R11/R15H Series （Optional）

Teach Pendant

Teach Pendant Model	IRTP200-LS-INT
cable length	5m/10m/20m/30m
Screen	Touch screen operation, function keys
IP rating	IP54

Expansion Card

Expansion Card Model	IRCB501-0016ETND-BD	IRCB501-1600END-BD	IRCB501-2ENID-BD	IRCB501-2PN-BD	IRCB501-FS-01-BD	IR-EC-CCLINK
Description	General I/O expansion card with 16 NPN outputs	General I/O expansion card with 16 inputs	2-channel differential input incremental encoder expansion card	PROFINET expansion card	Safety function expansion card	CC-LINK expansion card
Matching controller	IRCB501 Series, IRCB501 High-protection Series					

Robot application scenarios

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Technical Data

Series		IR-R4H		IR-R7H				IR-R10H			IR-R15H			
Model		IR-R4H-54S-INT	IR-R4H-54S-2SV-INT	IR-R7H-70S-INT	IR-R7H-70S-2SV-INT	IR-R7H-90S-INT	IR-R7H-90S-2SV-INT	IR-R10H-120S-INT	IR-R10H-120S-2SV-INT	IR-R15H-145S-INT	IR-R15H-145S-2SV-INT	IR-R20H-120S-INT	IR-R20H-120S-2SV-INT
Structural style		Vertical axis cascading structure											
Number of Axes		6 axes											
Maximum reach (mm)		545.7		722.3		911.9		1201.2		1455		1218	
Repeatability (mm)		±0.02		±0.015		±0.02		±0.025		±0.04		±0.04	
Maximum Load(kg)		4		7		7		10		15		20	
IP rating		IP40 (IP67 optional)											
Max.Speed	J1 (° /s)	450		420		336		240	240	260	260	260	260
	J2 (° /s)	460		336		280		180	180	240	240	240	240
	J3 (° /s)	520		487.5		390		330	330	260	260	260	260
	J4 (° /s)	560		550		550		470	470	470	470	470	470
	J5 (° /s)	560		438		438		438	438	450	450	450	450
	J6 (° /s)	900		764.7		764.7		764.7	764.7	705	705	705	705
Maxi-mummotion range	J1 (°)	±170		±170		±170		±170	±170	±170	±170	±170	±170
	J2 (°)	-120~+110		-135~+80		-125~+80		-135~+100	-135~+100	-160~+90	-160~+90	-160~+70	-160~+70
	J3 (°)	-65~+195		-70~+190		-70~+190		-66~+210	-66~+210	-76~+210	-76~+210	-76~+200	-76~+200
	J4 (°)	±190		±190		±190		±190	±190	±190	±190	±190	±190
	J5 (°)	±120		±120		±120		±125	±125	±140	±140	±140	±140
	J6 (°)	±360		±360		±360		±360	±360	±360	±360	±360	±360
Allowed wrist torque	J4 (N·m)	8.86		16.9		16.9		18.59	18.59	26.5	26.5	35	35
	J5 (N·m)	8.86		16.9		16.9		18.59	18.59	26.5	26.5	35	35
	J6 (N·m)	4.9		9.4		9.4		9.8	9.8	12	12	20	20
Allowed wrist inertia	J4 (N·m)	0.2		0.49		0.49		0.6	0.6	0.9	0.9	1	1
	J5 (N·m)	0.2		0.49		0.49		0.6	0.6	0.9	0.9	1	1
	J6 (N·m)	0.067		0.15		0.15		0.2	0.2	0.3	0.3	0.4	0.4
Customer connections	signal lines (30V 0.5A)	12	20	17	42	17	42	12	42	24	41	24	41
	Built-in solenoid valve	0	2	0	2	0	2	0	2	0	2	0	2
	Air(, 0.59 (Mpa)	Φ4 mm × 4(Direct)	Φ4 mm × 1(Direct)	Φ4 mm × 4(Direct)	Φ4 mm × 1(Direct)	Φ4 mm × 4(Direct)	Φ4 mm × 1(Direct)	Φ4 mm × 4(Direct)	Φ4 mm × 1(Direct)	Φ6 mm × 4(Direct)	Φ6 mm × 1(Direct)	Φ6 mm × 4(Direct)	Φ6 mm × 1(Direct)
Operating conditions	Ambient temperature (° C)	0~45											
	Relative humidity	5% to 95% RH (non-condensing)											
	Maximum Temperature Gradient (° C/min)	1.5											
Shipment Conditions	Ambient temperature (°C)	0~45											
	Relative humidity	≤ 95% RH,non-condensing											
Storage Conditions	Ambient temperature (°C)	-10~55											
	Relative humidity	≤ 95% RH,non-condensing											
Cable		Stand:5m optional: 3m/10m/15m/High Flexible(3m/5m/10m/15m)											
Weight		24.5kg	25kg	31kg	31.5kg	33kg	33.5kg	50kg	50.5kg	155kg	155.5kg	150kg	150.5kg
Controller		IRCB5016KD-INT											
Mounting mode		Floor mounted/Ceiling mounted/Wall mounted											
Certification		Kcs, KC, CE, cSGSus, FCC, Safety Function, cleanliness Class4											